The HSIT Program: Coherence, Arithmetic, and the Riemann Hypothesis

A corrected survey of the Farey-operator program, its finite lattice companion, its obstructions, and its open boundary

Author: Micah Blumberg

Self Aware Networks Research Institute

https://selfawarenetworks.com

July 2026

Abstract

The motivating problem is simple to state and difficult to solve. Can an explicit operator on arithmetic data expose structure relevant to the Riemann Hypothesis rather than merely translate a classical criterion into new notation? This survey records what the HSIT program has established, what later audits corrected, and what remains open.

The surviving mathematical core is substantial but bounded. Denominator damping turns the infinite-degree Farey graph into a bounded operator family exactly when `Re s > 1/2`. At finite level, the constant kernel vector and a position unitary recover the classical Mertens exponential sum. The associated commutator identity is exact, but a later algebraic audit established that it is a universal kernel-projection identity: the positive spectrum cancels. It therefore gives an exact operator-language restatement of the classical Mertens criterion, not a new proof mechanism by itself. The canonical operator has a trace-class commutator, a first positive eigenvalue bracketed between orders `1/(N^4 log N)` and `1/N^4`, and rigid global traces that exclude broad proposed carriers while leaving kernel-sensitive, shell-local, and off-diagonal observables open. The real Ford-damped family has a proved existence boundary at `s=1/2`, an explicit Hurwitz diagonal, and a fixed-level rescaled boundary moment equal to the classical totient ratio `ζ(2w-1)/ζ(2w)`. No theorem interpolates that boundary identity to the Mertens endpoint.

The SIT-inspired finite lattice companion has also been corrected. It proves a structural correspondence of restricted gauge orbits, fixed-magnitude factorization for phase / link events, and quartic pinning bounds. It does not preserve the action or Gibbs weights of the toy model, and transfer of the toy abundance constants remains open. The computational package verifies declared finite identities and negative controls; its trained Tier-3 protocol is unimplemented and reports no trained result.

The program's durable contribution is thus twofold: a concrete family of arithmetic graph operators with several proved boundaries, and a source-faithful method for correcting an AI-assisted research program without erasing its valid results. No result here proves the Riemann Hypothesis, HSIT, or physical Super Information Theory.

Scope and evidence boundary

Record continued: doi:10.5281/zenodo.21231894

1. The problem before the terminology

The Riemann Hypothesis is equivalent to square-root cancellation in the Mertens function `M(N)`. Rephrasing that criterion is easy; obtaining new analytic leverage is hard. An operator program must therefore answer three questions:

1. Is the operator actually defined on the proposed infinite carrier?

2. Which of its observables retain arithmetic information?
3. Does any retained observable depend on structure beyond the classical Mertens sum itself?

The HSIT program began from a physics-motivated ansatz and then submitted that ansatz to a source and value-map audit. The name is historical. The results below stand or fall on explicit graph, operator, and finite-lattice definitions, not on a physical interpretation.

The audit produced both mathematics and reversals. It retained a weighted Farey Laplacian, proved its exact summability boundary, found useful finite observables, and published obstructions. It also removed an undefined dense normalizer, corrected a gap lower bound, reclassified old numerical exclusions, showed that the apparent spectral saturation law is universal at the kernel-projection level, and retracted an action-preserving lattice embedding. Those corrections are part of the result.

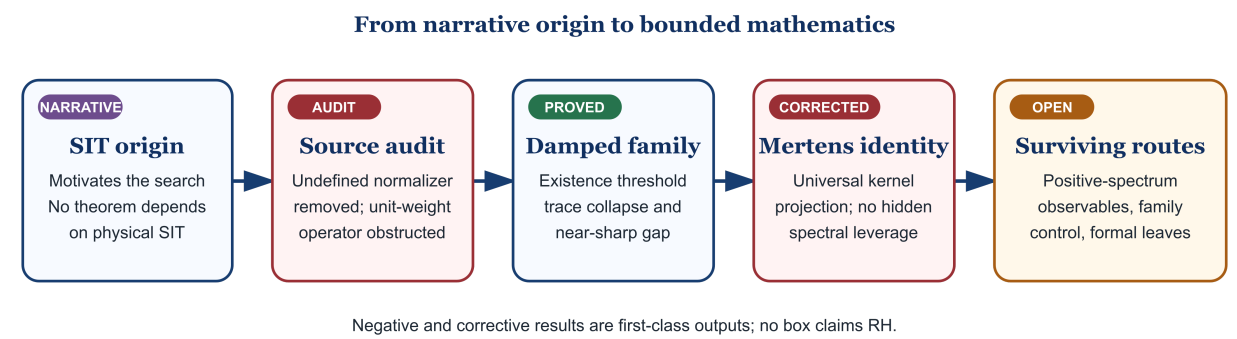

Figure 1: The corrected program arc

Figure 1. The corrected program arc. Narrative provenance motivates the search; only defined objects and proved statements carry mathematical weight.

2. Objects, notation, and evidence classes

Let F_N be the reduced fractions p/q in $[0,1)$ with $q \leq N$, and let $m_N=|F_N|$. Two reduced fractions a/b and c/d are Farey neighbors when $|ad-bc|=1$. The infinite Farey graph is G, and G_N is its finite level-N truncation.

For a neighbor edge $u \sim v$, with denominators q_u, q_v, the program studies the real or complex weighted family

$$w_s(u,v) = (q_u q_v)^{-2s}, \qquad \sigma = \operatorname{Re} s. \tag{1}$$

The associated magnetic graph Laplacian is $L(s)$; $L=L(1)$ is the canonical member and $L_N(s)$ its finite truncation. At finite level, ϕ_0 denotes the normalized constant kernel vector, U is the position unitary $(U\psi)(\rho)=e^{2\pi i\rho}\psi(\rho)$, P_0 projects onto the kernel, and L_N^+ is the Moore-Penrose pseudoinverse.

Statements use six evidence classes:

- **proved:** derived in the owning manuscript;
- **finite verification:** deterministic replay of a declared finite object;
- **measured:** descriptive numerical output at stated sizes;
- **conjecture:** a mathematical proposition not proved;

- **program:** a proposed route with open obligations;
- **narrative:** source history that motivates but does not prove.

No movement between those classes is implicit.

3. From the historical ansatz to a defined operator

The historical expression HSIT=V(n)+A(ij)/A(n×n) was not a theorem-ready definition. The audit asked whether each symbol supplied a value map. The vertex term and neighbor coupling could be made explicit; the dense `A(n×n)` term had no rule assigning values to its entries. It was removed rather than guessed.

The remaining unit-weight graph still failed: every Farey vertex has infinitely many neighbors, so the unit-weight quadratic form diverges on finitely supported vectors that are nonzero at that vertex. Denominator damping repairs this problem. The corrected companion papers prove that `L(s)` is bounded for σ`>1/2` and fails the stated form construction for $\sigma \leq$`1/2`. For real σ`>1/2`, it is positive and self-adjoint; the canonical commutator is trace class.

For real `s`, Ford circles supply an exact geometric dictionary. The circle associated with `p/q` has radius `1/(2q^2)`, and the depth below the cusp horoball is `2 log q`. Consequently the canonical edge weight is exponential cusp damping:

$$(q_u q_v)^{-2} = \exp[-d(B_\infty, B_u) - d(B_\infty, B_v)]\,. \tag{2}$$

The geometry explains the weights after the summability repair; it was not an independent derivation of the arithmetic endpoint.

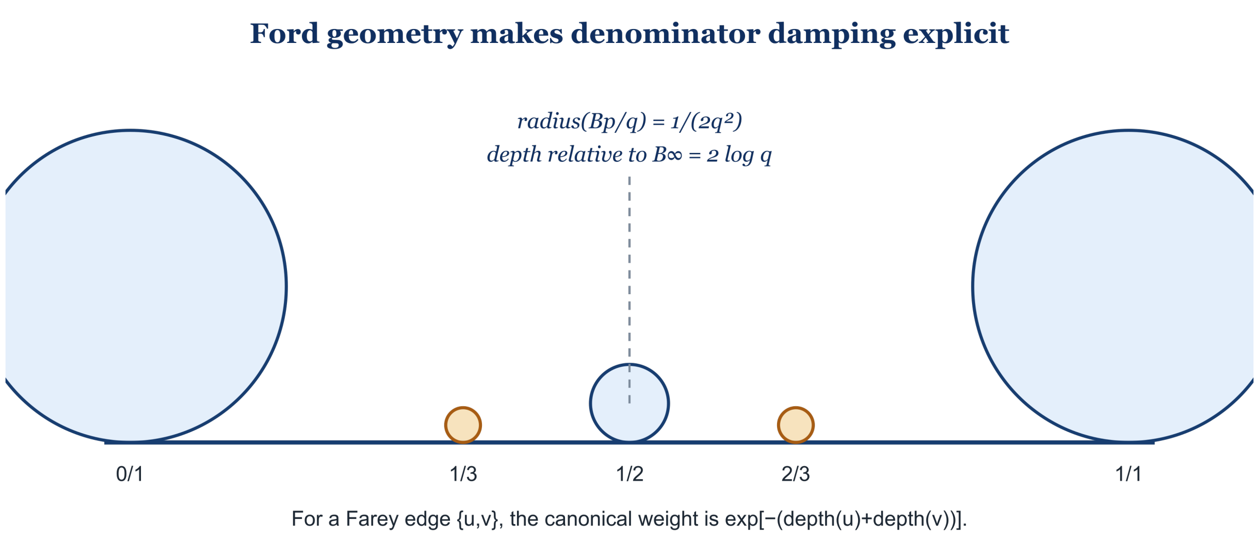

Figure 2: Ford-circle dictionary

Figure 2. Farey adjacency, Ford-circle size, and denominator depth. The diagram is explanatory; Equation [2] and the cited companion proof are the authority.

4. The arithmetic identity and its decisive correction

Grouping the constant-mode exponential sum by denominator yields Ramanujan sums, hence

$$\langle \phi_0, U\phi_0 \rangle = \frac{1}{m_N} \sum_{\rho \in F_N} e^{2\pi i \rho} = \frac{M(N)+1}{m_N}. \tag{3}$$

This identity is classical arithmetic carried by the constant mode. It does not depend on the positive eigenvalues, edge weights, or gap.

The program originally presented the next identity as a spectral conservation law. The corrected operator algebra gives a stronger and more limiting result. For any finite-dimensional self-adjoint `L` with `Lϕ_0=0`,

$$L^+[U, L]\phi_0 = -(I - P_0)U\phi_0. \tag{4}$$

Therefore

$$|\langle \phi_0, U\phi_0 \rangle|^2 + \|L^+[U, L]\phi_0\|^2 = 1. \tag{5}$$

Equations [4]-[5] depend on the kernel projection, not the positive spectrum. Replacing the positive eigenvalues while preserving the kernel leaves them unchanged. This is not a minor disclaimer: it determines what the identity can and cannot accomplish.

Define the deficit

$$\delta(N) = 1 - \|L_N^+[U, L_N]\phi_0\|^2 = \left(\frac{M(N)+1}{m_N}\right)^2. \tag{6}$$

Since `m_N=(3/π^2)N^2+O(N log N)`, the classical Mertens criterion becomes

$$\text{RH} \quad \Longleftrightarrow \quad \delta(N) = O_\varepsilon(N^{-3+\varepsilon}) \text{ for every } \varepsilon > 0. \tag{7}$$

This equivalence is exact. It is also circular as a proof strategy unless an additional Farey observable controls the kernel coefficient without merely restating a bound on `M(N)`. The earlier phrase “saturation law” remains a useful description of Equation [7], but not evidence of positive-spectrum leverage.

5. What the spectrum still contributes

The projection correction does not erase the operator theory. The canonical commutator is trace class, and the first positive eigenvalue satisfies

$$\frac{c}{N^4 \log N} \leq \lambda_2(L_N) \leq \frac{C}{N^4} \qquad (N \geq 2), \tag{8}$$

for unspecified absolute positive constants. The lower bound follows the smaller-parent canonical-path argument; the upper bound uses an explicit test vector. Removing the logarithm remains open.

The current finite package replays the declared `N=80` artifact and computes the same-system diagnostic at `N=20,40,60,80`. These values are descriptive. Older regression exponents, shell fits, and localization labels are preserved as historical discovery routes, not promoted as theorem evidence. In particular, the present package does not prove a universal localization law, pairing law, or asymptotic mobility transition.

Global traces have a different obstruction. For fixed regular test functions, the damped family has rigid trace growth: an extensive g(0)m_N contribution plus a bounded remainder, while the relevant Farey discrepancy is O(N\log\N)=o(m_N). Thus an affine trace identity must cancel its extensive term. Complete exclusion of the remaining cancelled branch requires the explicit optional hypotheses named H2 or H3 in the obstruction paper. The theorem does not cover N-dependent test functions, kernel-sensitive pairings, shell-local observables, or off-diagonal traces.

6. The real parameter family and the boundary

The Ford-damped family supplies a genuine parameter-space result. The exact existence boundary is Re s=1/2; the diagonal admits a Hurwitz-zeta description and entrywise continuation. At fixed N, the current boundary theorem gives the finite moment Σ_{q≤N}ϕ(q)q^{-2w}. Only after that real boundary limit is taken does N→∞ give, locally uniformly for Re w>1, the classical totient Dirichlet series:

$$\sum_{q\geq 1} \frac{\varphi(q)}{q^{2w}} = \frac{\zeta(2w-1)}{\zeta(2w)}. \tag{9}$$

This ratio also appears among Eisenstein-series coefficients. The equality is a classical identity; the program's result is the way it arises from the stated fixed-level boundary construction. It does not prove a spectral equivalence with the automorphic Laplacian.

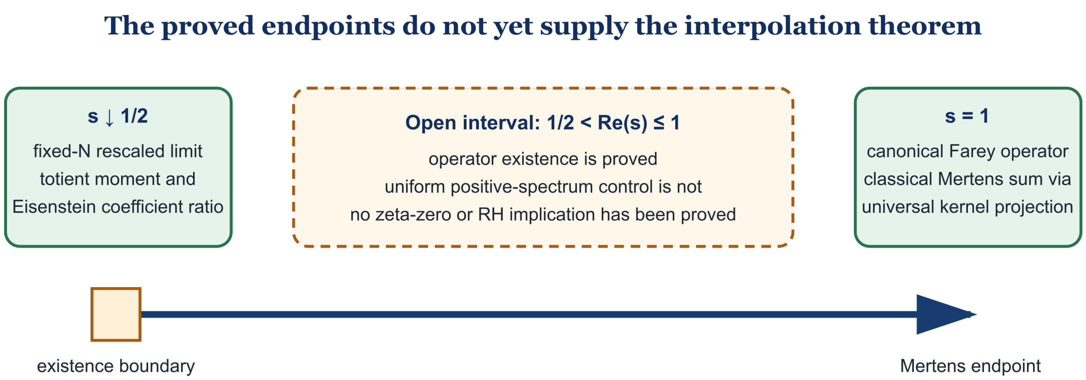

Figure 3: The real Ford-family segment

Figure 3. Proved endpoint facts and the open middle. The left side is a fixed-level rescaled boundary result; the right side contains the canonical operator and Mertens expectation. Uniform positive-spectrum control across the segment remains open.

This corrects an earlier interpolation narrative. Two meaningful endpoint facts do not by themselves define a bridge. Any surviving route must identify an observable that (i) depends genuinely on the positive spectrum or other non-kernel structure, (ii) is controlled uniformly as s approaches the boundary, and (iii) recovers arithmetic information without importing the desired Mertens estimate.

7. The finite lattice companion

The SIT-inspired lattice paper is an auxiliary mathematical model, not a proof of physical SIT. Its corrected finite gauge action admits a structural restricted-sector correspondence: on a finite simply connected square box, restricted phase / link configurations modulo gauge are in bijection with declared edge assignments, preserving plaquette holonomies and observables that depend only on those holonomies.

The correspondence is not an action identity. At unit magnitude an additional nonconstant edge functional remains, so action values, Gibbs weights, energy ordering, partition functions, and spectra are not inherited from the toy model. The former "exact embedding" claim is retracted.

At a fixed radial profile `R`, the paper proves factorization for phase / link events:

$$Z(W;R) = e^{-\lambda \sum_x (R_x^2-1)^2} \left(\prod_x z_x(R_x) \right) Z_{\text{phase}}(W;R). \tag{10}$$

It also proves quartic pinning and counting bounds for the displayed action. Those theorems do not imply unconditional independence after integrating over `R`, do not give a product factorization for total-action windows, and do not transfer the toy-model abundance constants. The remaining lift is therefore a family of explicit comparison problems rather than a completed theorem.

8. Computation and reproducibility

The computational companion separates finite replay from scientific extrapolation.

- The deterministic tier verifies the finite kernel, commutator orthogonality, Mertens trace, declared finite certificates, and negative mutations.
- The baseline tier reports one-system descriptive distances and explicitly does not infer an ensemble law, universality class, or zeta-zero match.
- The trained Tier-3 design requires held-out targets, fake-target controls, equal budgets, pinned seeds, run ledgers, and certificate checks.

Tier 3 is currently unimplemented. Its result array is empty. No trained-model result is reported. This closed result is preferable to a placeholder because it tells the reader exactly what exists and what does not.

Finite checkers test the objects they instantiate. They do not prove infinite-volume bounds, asymptotic laws, HSIT, SIT / Riemann, or RH.

9. Counterexample and bypass governance

Two later companion papers demonstrate why a research program must preserve failed bridges.

Paper 10 gives a Lean-certified endpoint-shape counterexample. It refutes a family-coverage bridge that had been treated as if it covered an arbitrary domain. Paper 11 then proves the target endpoint through a direct mediator constructor. The second result is a bypass, not a repair of the refuted classifier.

This distinction prevents a common loss of information:

- the counterexample remains true;
- the direct constructor remains true;
- the new proof does not retroactively validate the old bridge.

The same rule governs the present survey. The universal projection correction does not erase the Farey operator, and the lattice action mismatch does not erase the structural orbit theorem. A successor should preserve both the valid result and the reason an earlier inference failed.

10. The surviving research program

The nearest open questions are now narrower and more informative.

1. Can one construct a kernel-sensitive or positive-spectrum observable that retains Mertens-scale information without reducing algebraically to Equation [3]?
2. Can the logarithm in the gap lower bound [8] be removed?
3. Which shell-local or off-diagonal traces evade the global-trace rigidity theorem and admit uniform estimates?
4. Does the real family admit a useful uniform theory as `s↓1/2`, with constants and modes of convergence stated explicitly?
5. Is there a noncircular relation between a positive-spectrum observable near `s=1` and the fixed-level boundary moment [9]?
6. For the auxiliary lattice model, can one prove transfer of one declared toy quantity after accounting for the nonconstant edge action and radial fluctuations?
7. Can the finite formal layer be extended in Lean without promoting a finite theorem to an infinite or RH statement?

A successful interpolation program must answer Questions 1 and 5. Merely placing the Mertens endpoint and the totient-ratio boundary in one diagram is not sufficient.

11. Governance and lessons

The work supports a concrete source-faithful workflow:

1. preserve the original version and its claimed route;
2. audit every symbol for a value map;
3. separate theorem, finite replay, measurement, conjecture, program, and narrative;
4. compile load-bearing finite identities into deterministic checks;
5. run negative mutations that would fail if the checker were vacuous;
6. publish obstructions and counterexamples as first-class results;
7. when a claim fails, recover the strongest valid narrower statement;
8. never let a bypass silently rehabilitate a refuted bridge.

This survey is itself a correction artifact. Version 1 overstated positive spectral content, retained superseded numerical exclusions, described a lattice correspondence as action-exact, and left source and result placeholders. Version 2 removes those defects while preserving the defined operator family, the classical arithmetic identity, the gap theorem, the boundary construction, the finite lattice theorems, and the program's historical genealogy.

The governance evidence comes from one program conducted by one author with AI assistance. It is a worked case, not a population-level demonstration that the workflow is universally sufficient.

12. Limitations

This survey proves no new theorem independent of its companion manuscripts. It reports their current final-preprint boundaries.

No claim of RH is made. The exact deficit equivalence is a restatement of the classical Mertens criterion, and the universal projection identity supplies no positive-spectrum mechanism by itself.

No claim of physical HSIT or SIT is made. SIT is historical motivation for the mathematical search. The finite lattice paper is an auxiliary model inspired by SIT and does not supply a faithful physical discretization.

The boundary moment is fixed-level and does not establish uniform operator convergence, a spectral equivalence with an automorphic Laplacian, or an interpolation theorem.

The finite computations are regression and descriptive checks at their stated sizes. Tier 3 is unimplemented. There is no trained-engine result.

Internal machine checks and adversarial reviews do not substitute for independent specialist review.

13. Version lineage and authorial provenance

The original survey and its companion papers were developed through a human-directed, AI-assisted process. Micah Blumberg supplied the originating SIT / HSIT research program, selected the objects and questions, directed the correction policy, and is the author. AI systems assisted with derivations, code, drafting, and adversarial review. All mathematical claims are limited to the displayed definitions and the owning companion manuscripts.

This successor is lossless in the sense relevant to scientific revision: it preserves the historical hypotheses and routes as history, retains every current valid theorem at its proved scope, records retractions rather than deleting them, and adds the later counterexample / bypass lesson. It does not represent superseded claims as current results.

14. Conclusion

The HSIT program did not prove the Riemann Hypothesis. It did produce a well-defined Ford-damped Farey operator family, an exact existence boundary, a classical Mertens carrier, a universal projection correction, trace and gap theorems, a fixed-level totient-ratio boundary construction, and a governed set of open questions. Its auxiliary lattice branch proved finite structural and factorization results while retracting action-level equivalence. Its computational branch now states plainly that no trained result exists.

The next mathematical advance cannot come from repeating the kernel identity. It must exhibit and control new information: a positive-spectrum, shell-sensitive, off-diagonal, or otherwise noncircular observable whose arithmetic content survives the proved obstructions. That is the program's current frontier.

References

Companion papers and program records

1. M. Blumberg, *Weighted Farey Laplacians: Bounded Operators, Finite Approximation, a Universal Projection Identity for the Mertens Function, and Near-Sharp Gap Bounds*, successor to Zenodo record 21231873, 2026. doi:10.5281/zenodo.21231873
2. M. Blumberg, *A Constructive Coherence-Tower Theorem for a U(1) Lattice Toy Model of Super Information Theory*, Zenodo record 21231875, 2026. doi:10.5281/zenodo.21231875
3. M. Blumberg, *The Riemann Hypothesis as a Saturation Law*, Zenodo record 21231877, 2026. doi:10.5281/zenodo.21231877
4. M. Blumberg, *Obstruction Theorems for Farey-Spectral Approaches to the Riemann Hypothesis*, successor to Zenodo record 21231879, 2026. doi:10.5281/zenodo.21231879

5. M. Blumberg, *Spectral Gap and Finite-Level Cusp-Depth Diagnostics of the Weighted Farey Laplacian*, successor to Zenodo record 21231881, 2026. doi:10.5281/zenodo.21231881
6. M. Blumberg, *An Auxiliary Finite Lattice Model Inspired by SIT*, successor to Zenodo record 21231885, 2026. doi:10.5281/zenodo.21231885
7. M. Blumberg, *The Ford-Damped Operator Family: The Boundary Re s=1/2, a Totient Moment, and an Eisenstein-Coefficient Comparison*, successor to Zenodo record 21231887, 2026. doi:10.5281/zenodo.21231887
8. M. Blumberg, *A Bounded Reproducibility Package for the H_SIT Program: Corrected Claims, Finite Regressions, and Open Controls*, successor to Zenodo record 21231891, 2026. doi:10.5281/zenodo.21231891
9. M. Blumberg, *A Lean-Certified Endpoint-Shape Counterexample in the Branch-C Farey Mediator Program*, Zenodo record 21231896, 2026. doi:10.5281/zenodo.21231896
10. M. Blumberg, *Bypassing a False Shape Bridge: A Lean-Certified Direct Mediator Constructor in the Branch-C Farey Program*, final-preprint companion, 2026.
11. M. Blumberg, *Super Information Theory*, Zenodo record 18133332, 2025. doi:10.5281/zenodo.18133332

Classical and adjacent literature

12. L. Báez-Duarte, "A strengthening of the Nyman-Beurling criterion for the Riemann hypothesis," *Atti Accad. Naz. Lincei*, 14, 5-11, 2003.
13. A. Beurling, "A closure problem related to the Riemann zeta function," *Proceedings of the National Academy of Sciences*, 41, 312-314, 1955.
14. J.-F. Burnol, "A lower bound in an approximation problem involving the zeros of the Riemann zeta function," *Advances in Mathematics*, 170, 56-70, 2002.
15. L. R. Ford, "Fractions," *American Mathematical Monthly*, 45, 586-601, 1938.
16. G. H. Hardy and E. M. Wright, *An Introduction to the Theory of Numbers*, Oxford University Press.
17. A. E. Ingham, "On two conjectures in the theory of numbers," *American Journal of Mathematics*, 64, 313-319, 1942.
18. H. Iwaniec, *Spectral Methods of Automorphic Forms*, 2nd ed., American Mathematical Society.
19. D. H. Mayer, "The thermodynamic formalism approach to Selberg's zeta function for PSL(2,Z)," *Bulletin of the American Mathematical Society*, 25, 55-60, 1991.
20. B. Nyman, *On the One-Dimensional Translation Group and Semi-Group in Certain Function Spaces*, thesis, Uppsala, 1950.
21. E. C. Titchmarsh, revised by D. R. Heath-Brown, *The Theory of the Riemann Zeta-Function*, 2nd ed., Oxford University Press, 1986.